



>> Beyond Touch << NeuroMotion: EMG-Based Mobile Control Sys-

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Project Overview

NeuroMotion is an IoT-based assistive system for individuals with motor impairments. It utilizes a MyoWare EMG sensor and an ESP32 microcontroller to detect muscle flexes. A single flex acts as a 'click', and a double flex acts as a 'swipe', sending commands via BLE to a custom Android application. Compared to expensive traditional assistive devices, NeuroMotion offers an affordable, wearable solution using open-source technology.

Limitations:

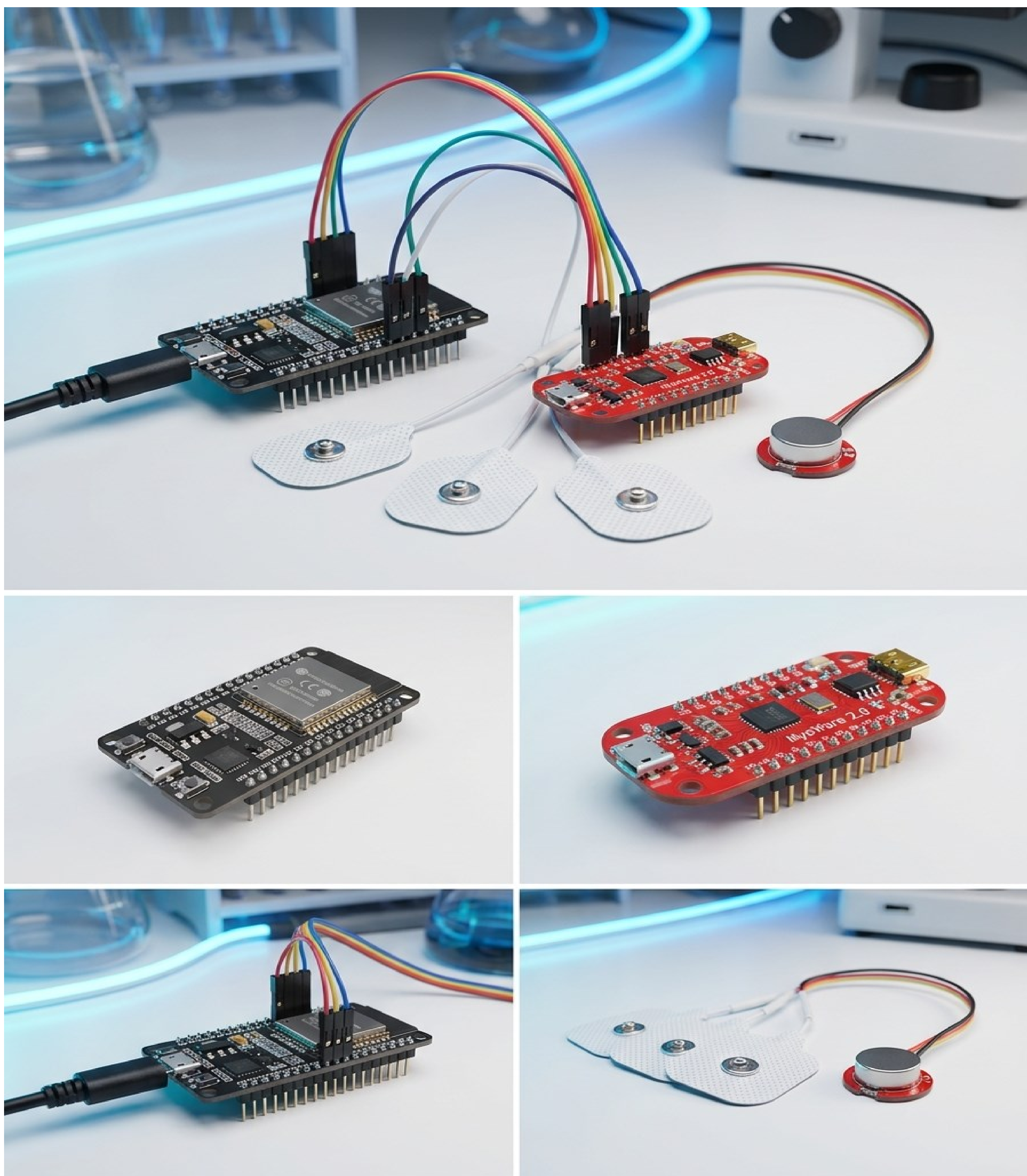
The current system relies on surface EMG, which can occasionally be affected by skin moisture and sensor placement precision. Additionally, it currently supports a limited number of gesture classifications.

Future Work:

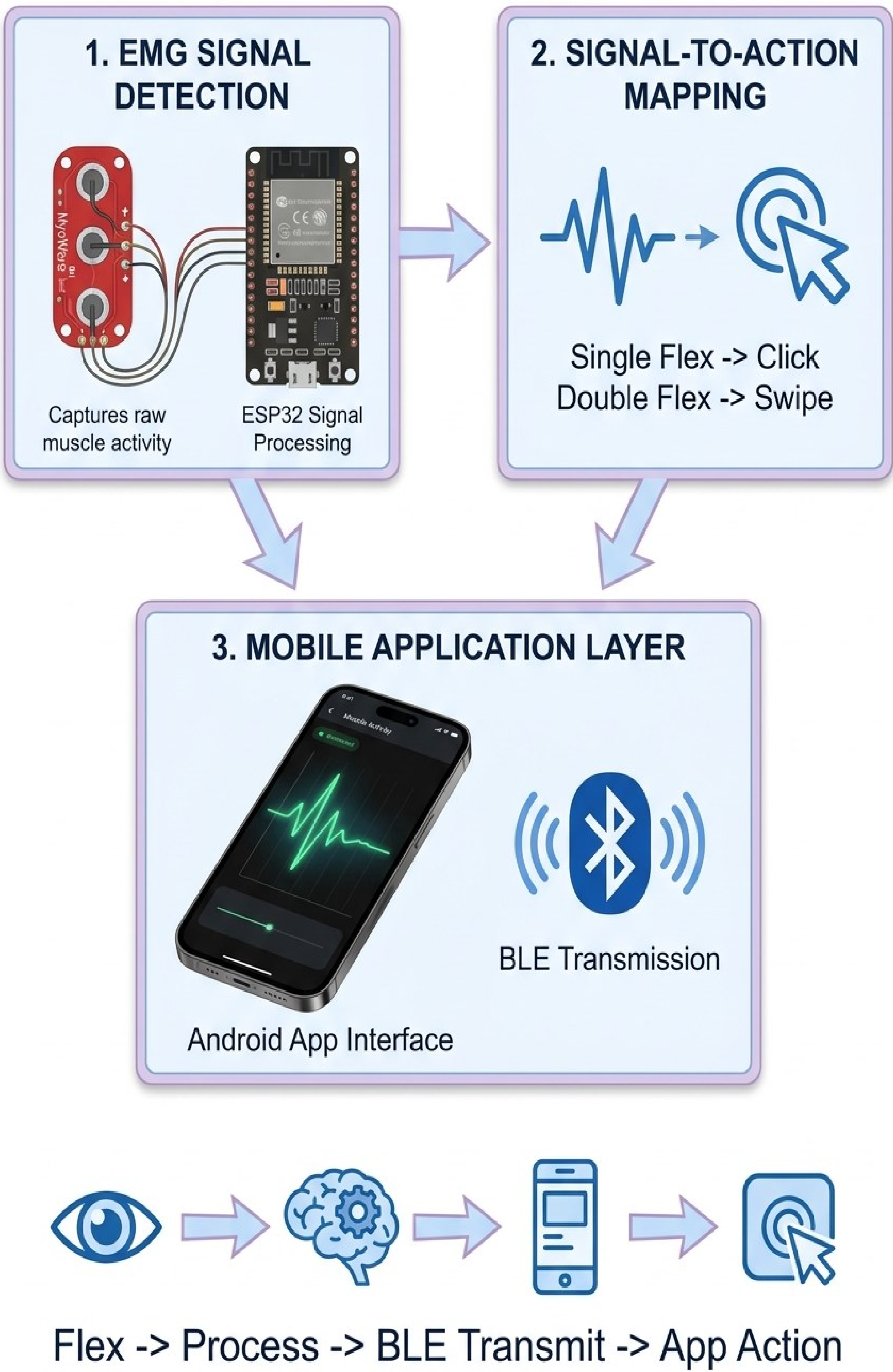
Future iterations aim to integrate machine learning algorithms for advanced gesture recognition (e.g., distinguishing between different finger movements) and deploying the system to control smart home appliances directly.

IoT Hardware Results

The system uses a wearable MyoWare 2.0 EMG sensor connected to an ESP32 microcontroller. The hardware successfully filters raw muscle signals and distinguishes between intentional flexes and resting states. A vibration motor provides real-time haptic feedback upon successful signal transmission via BLE to the Android app.



Proposed Framework Diagram



Mobile Application Results

